Journal of Materials Research and Technology 30 (2024) 4777–4785

Contents lists available at ScienceDirect

Journal of Materials Research and Technology

journal homepage: www.elsevier.com/locate/jmrt

Effect of welding current and speed on solidification cracking susceptibility in gas tungsten arc fillet welding of dissimilar aluminum alloys: Coupling a weld simulation and a cracking criterion

Fatemeh Bodaghi [a], Mojtaba Movahedi [a,*], Suck-Joo Na [b], Lin-Jie Zhang [b], Amir Hossein Kokabi [a]

[a] Department of Materials Science and Engineering, Sharif University of Technology, PO Box 9466-11365, Azadi Ave., Tehran, Iran
[b] State Key Laboratory of Mechanical Behavior for Materials, Xi'an JiaoTong University, Xi'an, 710049, China

ARTICLE INFO

Handling editor. L Murr

Keywords:
Simulation
Gas tungsten arc welding
Solidification cracking criterion
Weldability test

ABSTRACT

In order to study the effect of welding current and speed on the solidification cracking susceptibility of weld, gas tungsten arc welding of a fillet weld with a cold filler wire was simulated. Simulation was conducted for dissimilar welding of Al–Cu–Mg to Al–Mg–Si aluminum alloys with an Al–Mg filler metal at the welding current range of 145–175 A and welding speed of 18–24 cm/min. The outcomes of simulations, namely temperature gradient, velocity of isotherms, and weld chemical composition obtained for various welding currents and speeds, were used as inputs for a solidification cracking model. The results showed that the dimensions of the weld profile obtained from the simulation exhibited good agreement with experimental results. However, a discrepancy of 10–23% was observed between experimental and simulated results in relation to one dimension of the weld profile, i.e. the distance between the fillet weld corners. This discrepancy was addressed to identify the underlying reasons. According to the model proposed in this study for solidification cracking of aluminum alloys, both higher welding current and higher welding speed led to a higher susceptibility to solidification cracking. This result was justified by considering factors such as secondary arm spacing of dendrites, isotherm velocity, backfill time, and chemical composition and viscosity of the weld pool. Weldability tests were employed to validate the model's performance for predicting the solidification cracking susceptibility. Thus, the optimal welding parameters can be chosen in accordance with these results and production demands.

1. Introduction

Al–Cu–Mg and Al–Mg–Si aluminum alloys are commonly used in various industries, especially the transportation industry, making the dissimilar welding of these alloys highly significant from a practical viewpoint. Gas tungsten arc welding (GTAW) is one of the most conventional techniques for welding of various aluminum alloys [1]. This process can be done with or without filler metal and its arc characteristics do not significantly depend on addition of the filler metal. However, GTAW of dissimilar aluminum welds may have some challenges, one of the most important ones being solidification cracking [2], which occurs due to tensile strain induced in the mushy zone during the solidification of the weld. The sources of this strain are solidification shrinkage and thermal contractions during solidification [3,4]. Clearly, solidification cracking reduces the quality of welded joints, making prevention essential. Overcoming this challenge and ensuring sound welds require careful selection of filler metal and optimization of welding parameters, such as speed and current [5,6]. However, finding the appropriate welding parameters typically necessitates numerous experimental tests to minimize the risk of solidification cracking and achieve desired joint quality. An alternative method for predicting solidification cracking is the utilization of developed numerical models to determine the solidification cracking susceptibility (SCS). There are some models in the literature for predicting the SCS [7]. Two of the most famous models, developed by Rappaz et al. [8] and Kou [9], are known as the RDG model and the Kou model, respectively. Kou [9] investigated weld solidification cracking by considering three factors along a grain boundary (GB) between two adjacent columnar dendritic grains: 1. detachment of grains from each other under tensile strain, leading to cracking, 2. growth of grains towards each other, resulting in the formation of a bridge to resist cracking, and 3. fluid flow along the GB to suppress cracking. Kou proposed that the most important factor affecting the SCS is $(d\sqrt{f_s}\ /dT)$ near the $f_s = 1$. A higher absolute value of

* Corresponding author.
E-mail address: m_movahedi@sharif.edu (M. Movahedi).

https://doi.org/10.1016/j.jmrt.2024.04.195
Received 7 March 2024; Received in revised form 19 April 2024; Accepted 22 April 2024
Available online 23 April 2024

Nomenclatures

G	Temperature Gradient
λ_2	Secondary arm spacing
k	Thermal conductivity
E	Welding voltage
T_0	Ambient temperature
σ	Stefan-Boltzman constant
r	Local coordinate
μ_0	Permeability of vacuum
G2	Universal function
D	Arc tip diameter
ρ_ρ	Argon plasma density
Re_0	Reynolds number
f_s	Solid fraction
β	Shrinkage factor
V_T	Isotherm velocity
γ	Surface tension
μ	Kinematic viscosity
η	Arc efficiency
I	Welding current
h_c	Convection coefficient
ε	Material emissivity
R	Arc radii
μ_m	Material permeability
H	Arc length
c	Sheet thickness
U_0	Average plasma jet speed
A_u	Upper plate contribution in weld metal
A_b	Lower plate contribution in weld metal
A_f	Filler metal contribution in weld metal
A_w	Weld metal area
g	Gravity

($d\sqrt{f_s}/dT$), indicates a slower growth rate, representing less resistance to cracking. However, Kou's model only considered the solid phase, while the presence of liquid backfilling strongly affects the SCS. In contrast, the RDG model takes into account both solid and liquid phases. According to the RDG model, the maximum pressure depression (Δp_{max}) between two adjacent dendrites is calculated by Equation (1).

$$\Delta p_{max} = \frac{180}{\lambda_2^2}\frac{(1+\beta)\mu}{G}\int_{T_s}^{T_L}\frac{E(T)f_s(T)^2}{(1-f_s(T))^3}dT + \frac{180}{\lambda_2^2}\frac{v_T\beta\mu}{G}\int_{T_s}^{T_L}\frac{f_s(T)^2}{(1-f_s(T))^2}dT \tag{1}$$

when,

$$E(T) = \frac{1}{G}\int f_s(T)\dot{\varepsilon}(T)dT \tag{2}$$

If the Δp_{max} exceeds the cavitation pressure, a cavity forms at the GB and can initiate a solidification crack. In other words, a higher Δp_{max} indicates greater SCS, as it may exceed the cavitation pressure.

The important point to consider is that, in order to enhance the accuracy of this model in predicting sensitivity to solidification cracking, it is necessary to determine parameters such as weld chemistry, thermal gradient at the solidification front, and velocity of isotherms as inputs to the model. For determining the aforementioned parameters under various welding currents and speeds, simulation of welds can be utilized [10–15].

According to the best knowledge of the authors, no attempt has been made thus far to study the SCS of fillet welds of dissimilar aluminum alloys using the coupling of weld simulation and the RDG solidification cracking model. Thus, the objective of the present study is to investigate the SCS of gas tungsten dissimilar fillet welds between Al–Cu–Mg and Al–Mg–Si aluminum alloys using the RDG model under various welding currents and speeds. To achieve this goal, first, GTAW of Al–Cu–Mg to Al–Mg–Si aluminum alloys with addition of the Al–Mg filler metal in a fillet weld-lap joint configuration is simulated using Flow-3D software. Then, the chemical composition of the weld metal (by identifying the contribution of base metals and filler metal in the weld pool), temperature gradient and velocity of isotherms obtained from the simulation at various welding currents and speeds are used to ascertain the weld physical properties and predict the SCS according to the RDG model. The model predictions are further validated through weldability tests.

2. Experimental setup

The automatic GTAW system used in this work is shown in Fig. 1. The lap joints were made using Al6061 aluminum alloy as the upper sheet and Al2024 aluminum alloy as the lower sheet, in dimensions of 50×100×3 mm and 100×100×3 mm, respectively, and with the overlap of 50 mm. The joints were fillet welded using an alternating current GTAW machine with a current balance of 50% negative and 50% positive. Argon gas with a purity of 99.99% and a flow rate of 15 lit/min was used as the shielding gas during welding. Al5183 wire with 1.2 mm diameter was used as the filler metal using automatic cold wire feeding system. Chemical compositions of the base metals and filler metal are given in Table 1. The sample nomenclature system and process parameters are listed in Table 2. The wire feeding rate was determined by primary experiments in accordance with the welding speed. The GTAW torch had an angle of 45^O with respect to the surface of the upper sheets. The tungsten electrode was positioned in 3 mm distance from both sheets.

The welding was started from a distance of 1.5 cm to the edge of the sheets and was continued to 1 cm from the end edge of the base sheets, in order to avoid the edge effect. Two high speed Phantom Miro M120 Charged-coupled device (CCD) camera were set to capture the arc images from the arc rear and arc side positions. Arc images were analyzed using Abel inversion and Fowler-Milne methods to find the arc characteristics.

Welded samples were cross sectioned and polished. Then, they were electro-etched using a 2.5% HBF_4 solution to reveal the microstructure of the welds and measure the secondary dendrite arm spacing. The remaining parameters for SCS prediction were determined through JMatPro software, accounting for the chemical composition of the welds as obtained from the simulation results. The SCS was then calculated within the solid fraction range of 0.87–0.94, according to results reported by Ref. [16].

In order to confirm the SCS results predicted by the RDG model, the weldability test based on the Soysal et al. [17] approach was carried out. All welding conditions were considered as previously described. The initial movement velocity of lower base sheet was set at 0.5 mm/s for 4 s and then, the velocity was reduced to 0.2 mm/s. The ratio of the crack length to the weld length (Normalized crack length), was considered as the SCS index.

3. Theoretical formulation

3.1. Weld configuration and mesh size

For the simulation, the joint configuration and position of the base sheets were completely similar to the welded samples.

In order to describe the weld pool, at least four meshes are necessary for droplet diameter [18]. It should be noted that although ref [18] pertains to the gas metal arc welding (GMAW) process, it was only used

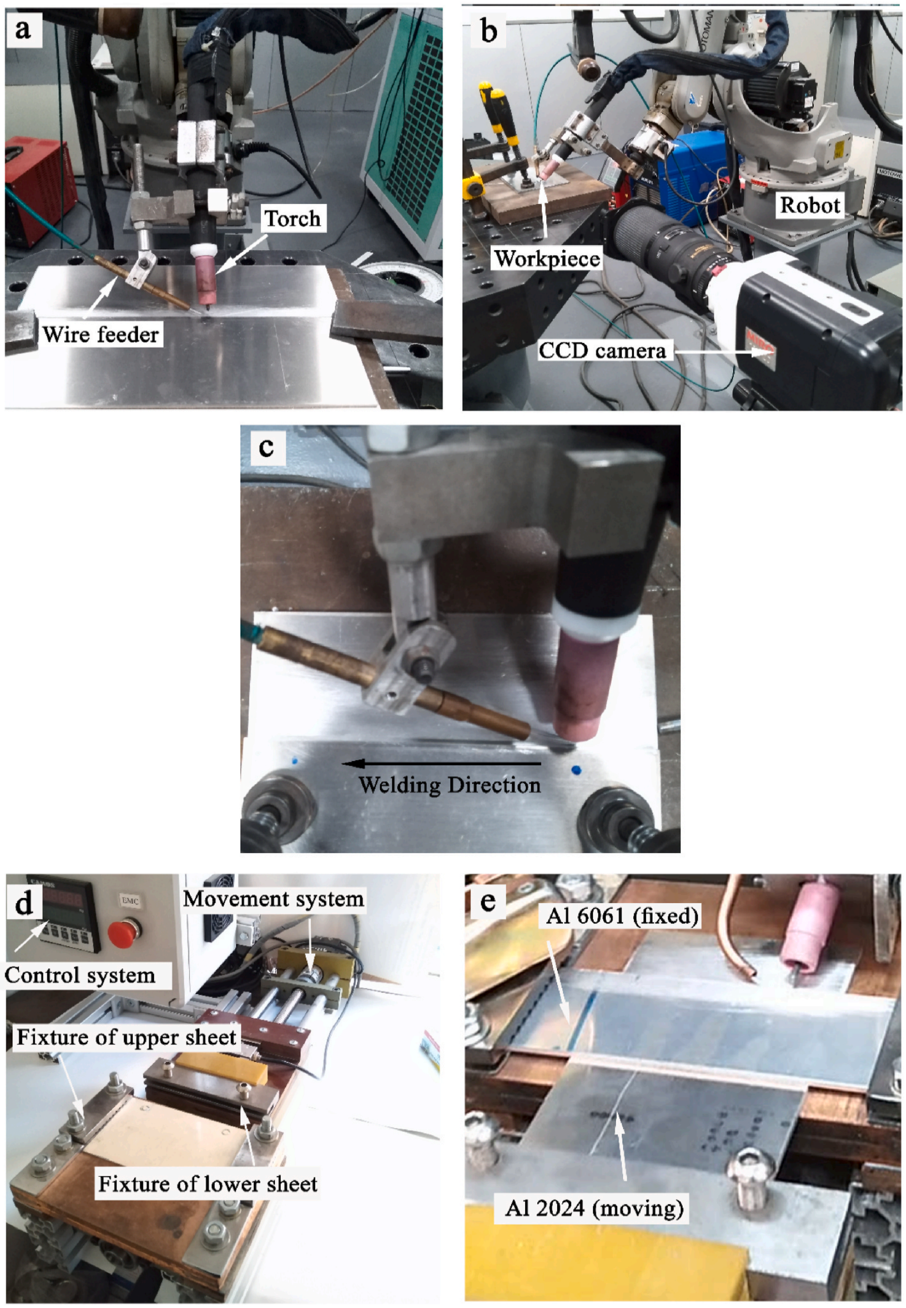


Fig. 1. a) Automatic GTAW system, b and c) CCD camera positions to capture rear and side images from the arc, d) weldability test device and e) arrangement of the base sheets in the weldability test.

Table 1
Chemical composition (in wt.%) of the alloys used as the base sheets and filler metal.

	Si	Cu	Mg	Mn	Fe	Cr	Zn	Ti	Ni
Al 2024	0.31	4.17	1.57	0.62	0.35	–	0.13	0.4	0.6
Al6061	0.62	0.33	1.5	0.12	0.28	0.23	0.12	0.3	–
Al5183	0.22	0.6	4.8	0.8	0.23	0.15	0.18	0.12	–

for determining the minimum number (size) of meshes required for simulation of droplet addition based on the droplet diameter. For other required parameters in this study, information related to the GTAW process was utilized. In the present work, as it is shown in Fig. 2, the cubic mesh with the size of 0.025 cm was chosen for simulation of the weld pool within $y = -2$ cm to $y = 2$ cm and for all x and z coordinates. To reduce the computing time and cost, larger meshes with the size of 0.05 cm was used for the regions far from the weld pool.

3.2. Boundary conditions and determination of parameters

To solve the mass conservation, energy, momentum, and Volume of Fluid (VOF) equations, the Flow-3D software was used. Surface was tracked by VOF method which is a kind of continuity equation, i.e. a

Table 2
GTAW parameters used for welding of the samples.

Sample name	Welding current (A)	Welding speed (cm/min)	Wire feeding rate (cm/min)
145 A- 18 cm/min	AC-145	18	90
160 A- 18 cm/min	AC- 160	18	90
160 A- 21 cm/min	AC- 160	21	95
160 A- 24 cm/min	AC- 160	24	100
175 A- 18 cm/min	AC-175	18	90

conversation of F value between zero and 1 [19].

On the top free surface, where the welding is done, arc heat flux (q_a), convection (q_{conv}) and radiation (q_{radi}) are the sources of heat input and heat loss. Therefore, the top free surface temperature was calculated using Equation (3).

$$k\frac{\partial T}{\partial \vec{n}} = q_a - q_{conv} - q_{radi} \tag{3}$$

$$q_{conv} = -h_c(T - T_0) \tag{4}$$

$$q_{radi} = -\sigma\varepsilon\left(T^4 - T_0^4\right) \tag{5}$$

Moreover, arc pressure (P), drag force (τ) and electromagnetic forces (EMFs) are applied from the top free surface. According to the transmission angle of 45^{O}, considered in this research, all arc equations such as arc heat flux (Equation (6)), arc pressure (Equation (7)), arc drag force (Equation (8)) and EMFs (Equation (9)), were redefined as below:

$$q_a = \frac{\eta EI}{2\pi R^2}\exp\left(-\frac{r^2}{2R^2}\right) \tag{6}$$

$$\begin{aligned} P(x') &= 0 \\ P(y') &= \frac{\mu_0 I^2}{4\pi^2 R^2}\exp\left(\frac{r^2}{2R^2}\right)\sin\left(\frac{\pi}{4}\right) \\ P(z') &= \frac{-\mu_0 I^2}{4\pi^2 R^2}\exp\left(\frac{r^2}{2R^2}\right)\cos\left(\frac{\pi}{4}\right) \end{aligned} \tag{7}$$

$$\tau = G2\left(\frac{r}{H}\right)\frac{\rho_p U_0^2}{\mathrm{Re}_0^{1/2}\left(H/D\right)^2} \tag{8}$$

In drag force equation (Equation (8)), G2, the universal function of drag force was determined according to the arc length/arc tip diameter (H/D), by Ref. [21].

As addition of melt to the weld pool was relatively continues and had no specific shape, it was difficult to measure the melt droplet diameter. Hence, the droplet diameter was determined using the volume of the consumed wire and the droplet frequency. After that, the droplet velocity was determined according to droplet velocity formulated by Ref. [22] for the short circuit metal transfer condition in GMAW. Temperature of the droplet was also considered equal to its liquidus temperature (T_l) when it was attached to the weld pool [23]. The physical properties of the base metals and filler alloys, obtained using the JMatPro software. In order to consider the effect of surface tension on the melt flow, the rate of surface tension variation with temperature ($\frac{d\gamma}{dT}$) was assumed to be constant and equal to 0.3013 mN/m.K. Physical properties of these alloys, especially thermal conductivity, are relatively different. The main issue here was that the Flow-3D software has been designed to investigate only one alloy during the process. It means that the properties of just one alloy can be defined for the software. This issue can limit the simulation of dissimilar welding. Therefore, since the most volume of the welding pool was filled with the filler alloy in the present work, the filler metal was considered to be the main alloy and the physical properties of this alloy were defined in the software. Then, properties of the base alloys were defined as a coefficient of properties of the filler alloy and the related routines of the Flow-3D software were changed for this purpose.

Since the welding current gradually increases from zero to the predefined maximum value, the arc radius also rises gradually. In other words, most of the time, the arc radius is less than the maximum value that the arc reaches it at the peak current. In this study, to reduce computational time and cost, alternating current was not utilized in the simulation. Instead, direct current with the same energy-input (heat-input), i.e., a current value equal to the root mean square of the maximum current in the alternating current (I_{RMS}), was considered. Therefore, there was practically no need to employ transient conditions for changes in electrical current within a cycle. Simulation domain and boundary conditions are shown in Fig. 2. The ‘C’ relates to continuative boundary condition which means a zero- normal derivative condition for all quantities, intended to represent a smooth continuation of the flow through the boundary.

$$\begin{aligned} F(x') &= \frac{-I^2\mu_m}{4\pi^2 rR^2}\exp\left(-\frac{r^2}{2R^2}\right)\left[1-\exp\left(-\frac{r^2}{2R^2}\right)\right]\left(1-\frac{z}{c}\right)^2\frac{x'}{r} \\ F(y') &= \frac{-I^2\mu_m}{4\pi^2 rR^2}\exp\left(-\frac{r^2}{2R^2}\right)\left[1-\exp\left(-\frac{r^2}{2R^2}\right)\right]\left(1-\frac{z}{c}\right)^2\frac{y'}{r}\cos\left(\frac{\pi}{4}\right) - \frac{I^2\mu_m}{4\pi^2 r^2 c}\left[1-\exp\left(-\frac{r^2}{2R^2}\right)\right]^2\left(1-\frac{z}{c}\right)\sin\left(\frac{\pi}{4}\right) \\ F(z') &= \frac{-I^2\mu_m}{4\pi^2 rR^2}\exp\left(-\frac{r^2}{2R^2}\right)\left[1-\exp\left(-\frac{r^2}{2R^2}\right)\right]\left(1-\frac{z}{c}\right)^2\frac{y'}{r}\sin\left(\frac{\pi}{4}\right) + \frac{I^2\mu_m}{4\pi^2 r^2 c}\left[1-\exp\left(-\frac{r^2}{2R^2}\right)\right]^2\left(1-\frac{z}{c}\right)\cos\left(\frac{\pi}{4}\right) \end{aligned} \tag{9}$$

Buoyancy force (f_g) was considered using *Boussinesq* approximation as the internal force according to equation (10).

$$f_g = -k(T - T_0)g \tag{10}$$

Arc efficiency for GTAW of aluminum alloys with alternating current was reported by Ref. [20] to be in the range of 0.63–0.85. Many tries were made to find the best η for the simulation and η = 0.8 was found to have the best match with the welded samples. To avoid procrastination, only the results obtained with η = 0.8 were reported here. As an asymmetric arc was considered, the arc radius (R) in each direction was defined according to the cell coordinate.

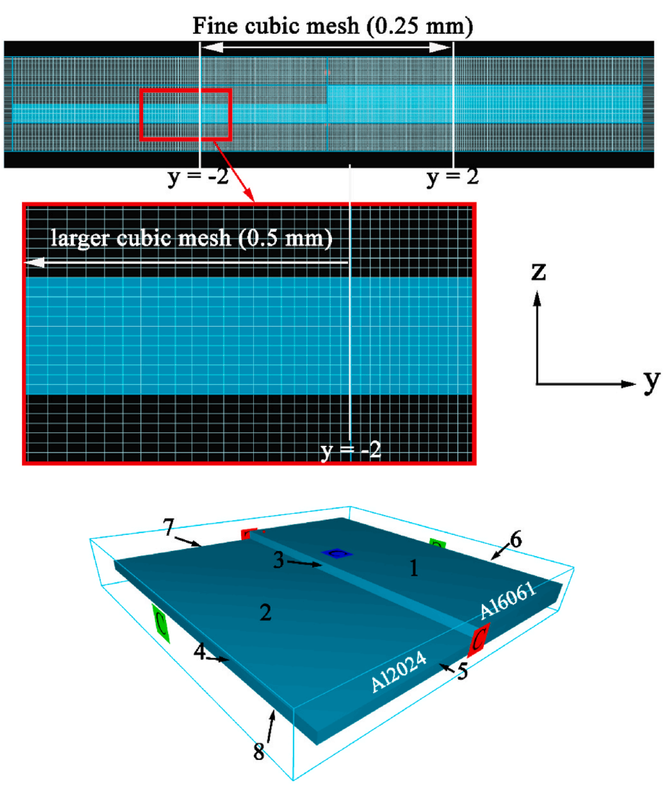
4. Result and discussion

4.1. Weld profile and validation of simulation results

Validity of the simulation results was verified by comparison to the experimentally welded samples. Fig. 3 exhibits the maximum temperature of each cell obtained from the simulation results along with the typical corresponding experimental cross sections. The maximum temperature at point (4.5,0,0) was compared with the simulation results at the same point, which is shown in Fig. 4(a). Simulation results showed that after 10 s from the start of welding, a steady-state condition is established for the welds, and the shape of the weld pool remains unchanged thereafter. Point (4.5, 0, 0) in the samples was in a steady state condition for all welding currents and speeds used in this study. Cheon et al. [21] considered the solidus temperature (T_S) as the criterion for the fusion zone boundary. However, formation of few content of liquid cannot significantly affect the microstructure which in turn cannot be depicted after etching the samples. Furthermore, T_l is not an accurate criterion for the fusion zone boundary. As the free surface of samples was predicted accurately by the simulation, an attempt was made to determine the temperature that best matches the fusion zone boundary in the experimental samples. It was found out that the temperature equal to $f_s = 0.7$ had the best match in all samples. Accordingly, if the maximum temperature of cell exceeded the T ($f_s = 0.7$), the cell was considered as the fused cell.

As expected, in both experimental and simulation cross sections, higher heat input due to the higher welding current and lower welding speed resulted in the wider and deeper weld pool which was also accurately predicted by the simulation. Fig. 4(b) is a graph displaying the average values of the weld cross-section dimensions, along with the simulation results. As shown in the figure, the simulation results were in good agreement with the experimental results, except for dimension A. The simulations underestimated the weld penetration in the base metals, particularly in the upper sheet, which in turn led to a reduction in dimension A in its results. This observation may be caused by the following reasons:

- Wu et al. [5] reported that the velocity of the droplet impingement influenced weld penetration in the base sheets. It should be mentioned that since the simulated process in this study involved GTAW along with the addition of filler wire, the addition of metal to the weld seam was considered as droplets detaching from the cold wire and entering the weld pool. In the present simulation, the droplet velocity was considered constant in each three directions (x, y,z). However, due to presence of different forces in the arc region such as EMFs and the way droplet is separated from the wire, the droplet velocity may not be constant in three directions.
- For welding of the samples, an alternating current was used. However, a constant current (I_{RMS}) was considered in the simulation. It should be noted that higher welding current leads to the wider arc, higher arc heat flux and arc forces which can influence the weld shape.
- In dissimilar welding, an alloy is formed in the weld pool which has different physical properties from the base and filler alloys. Moreover, there is not enough and accurate data for high temperature properties of the base alloys, filler metal and the alloy formed in the



Face No.	1	2	3	4 - 8
Boundary condition	A/V/R	A/V/R	A/V/R	V

Fig. 2. Mesh Size, domains and boundary conditions for the simulated fillet weld-lap joint (A: arc heat input and arc forces, V: convection, R: Radiation).

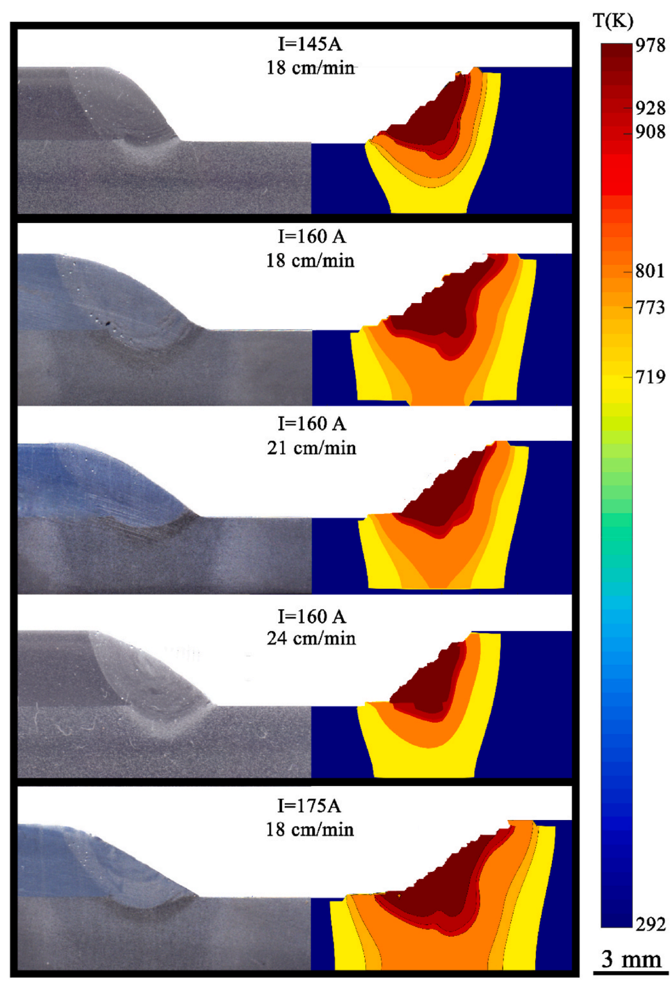


Fig. 3. Maximum cell temperatures of the simulation domains along with the cross section of the corresponding experimental weld.

weld pool. Based on what has been mentioned, a difference between the simulation and experimental results is expected [24].

4.2. Solidification cracking susceptibility of weld

In the present work, the simulation results were used to evaluate the SCS of welds according to the RDG model. For this purpose, physical properties of the welds were needed. The results obtained from simulation were employed to attain the required properties instead of having to weld numerous samples. All of physical parameters depend on the chemical composition of the weld. In dissimilar joining with filler metal, weld dilution governs the chemical composition of the welds and can be calculated as it is shown in Fig. 5(a), using the chemical compositions of the base metals and filler metal provided in Table 1. Hence, the weld dilution calculated from the simulation was utilized to determine the chemical composition of the welds (Table 3). As mentioned before, the secondary arm spacing (λ_2) was measured from weld cross sections. A typical weld microstructure is shown in Fig. 5(b) and measured λ_2 values are shown in Fig. 5(c). Moreover, temperature gradient (G) and velocity of the isotherms during the welding were determined from the simulation results for point (x,0,0) which are shown for a sample weld in Fig. 5 (d) and (e). Then, $f_s(T)$, viscosity and density of the welds were determined by JMatPro software given the weld chemical composition.

The maximum pressure depression obtained from the RDG model for all five cases is given in Fig. 6(a). Given the values of ΔP_{max} (directly related to the SCS) presented in Fig. 6, a higher welding current at a constant welding speed led to a higher SCS. This relationship was also confirmed by the results of the weldability test in Fig. 6 (b,c,f). This increase in SCS can be attributed to the higher weld dilution resulting from enhancement of the welding current, which in turn, leads to a higher copper (Cu) content in the weld. For example, the copper content increased from 1.29 wt% in the 145 A- 18 cm/min weld to 2.4 wt% in the 175 A- 18 cm/min weld (Table 2). Copper, being a heavy element, increases the density and viscosity of the melt, making back filling more difficult. Moreover, the length of the liquid channel (which can be calculated as the BTR divided by G) increases with the enhancement of welding current, going from 0.5 mm in the 145 A- 18 cm/min weld to 1.22 mm in the 175 A- 18 cm/min weld. A longer liquid channel between dendrites again makes backfilling more difficult and raises the SCS. In addition, with an increase in copper content in the weld metal, the likelihood of its segregation during non-equilibrium solidification increases, which can further enhance the susceptibility to cracking. On the other hand, an increase in welding current, and consequently, an increase in heat input, may also lead to higher thermal stresses on the weld metal. This phenomenon can encourage solidification cracking.

According to the results of the SCS model shown in Fig. 6(a) a higher welding speed led to a higher SCS. As the welding speed varies, the following factors affect the SCS:

(i) A higher cooling rate of weld metal due to higher welding speed leads to lower secondary arm spacing of dendrites, making the back filling difficult and increasing the SCS
(ii) Increasing the welding speed results in a higher velocity of isotherms (V_T), with values ranging from approximately 3.1 mm/s for the lowest welding speed to about 4.2 mm/s for the highest welding speed investigated in this study. This velocity has a direct correlation with SCS as predicted by the solidification cracking model (Equation (1)).
(iii) Increased welding speed leads to a reduced temperature gradient, shorter backfill time, and a larger zone prone to cracking, consequently raising SCS [25].
(iv) A higher welding speed reduces the weld dilution and thus, influences the weld pool chemical composition. As explained earlier, reducing the heavy elements (such as Cu in this work) can decrease the SCS.

Among the mentioned factors, the first three indicate that an increase in welding speed leads to an increase in SCS. However, the fourth factor suggests a decrease in SCS with an increase in welding speed. Hence, according to the presented model results, the factors increasing sensitivity to solidification cracking are predominant.

The weldability test results in Fig. 6(c–e) show that while higher welding speeds led to increased SCS at low and medium welding speeds (i.e. 18 and 21 cm/min), the SCS was again reduced at the high welding speed (24 cm/min). This discrepancy between the prediction of the SCS model and the result of the weldability test at high welding speeds can be attributed to two factors: First, the developed model is solely based on the microstructural aspects, and it does not take into account the mechanical stress aspects influencing the SCS with variation of the welding speed. For instance, a higher welding speed reduces the duration of stress exposure in the weld metal and relocates regions of compressive forces from areas away from the heat source to the mushy zone, resulting in a decreased SCS [25]. However, the result of the weldability test is governed by both the microstructural and mechanical aspects. Second,

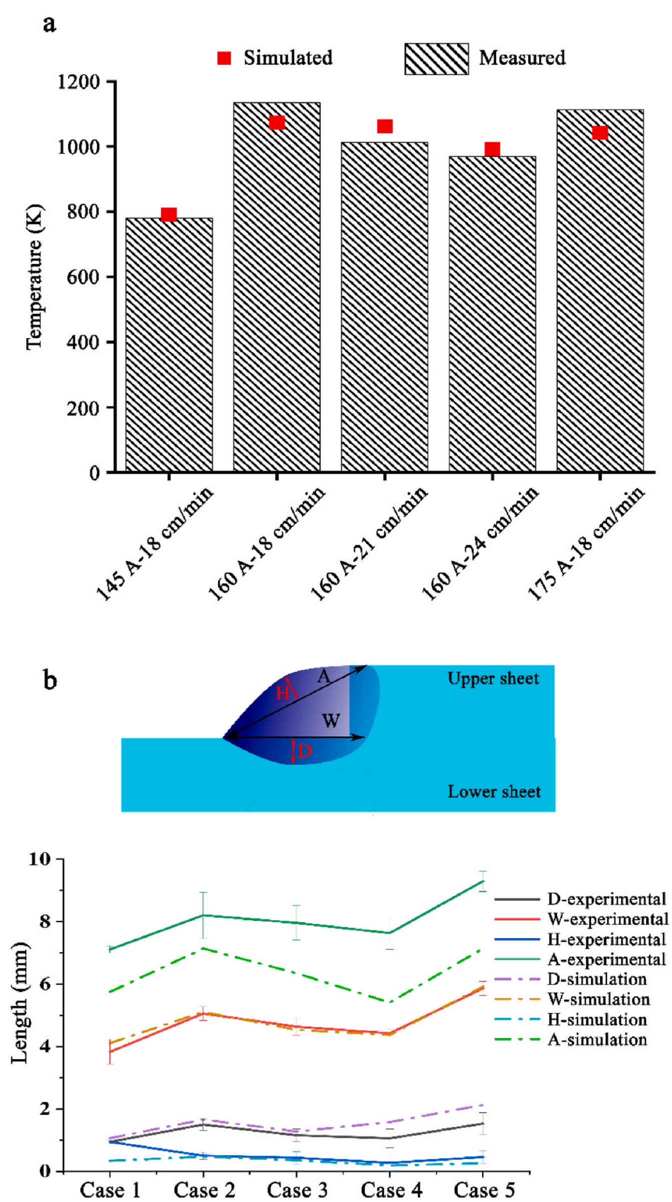


Fig. 4. a) Maximum temperature at point (4.5,0,0) for simulated and welded samples, b) Dimensions of the weld profile obtained from the experiments and simulations.

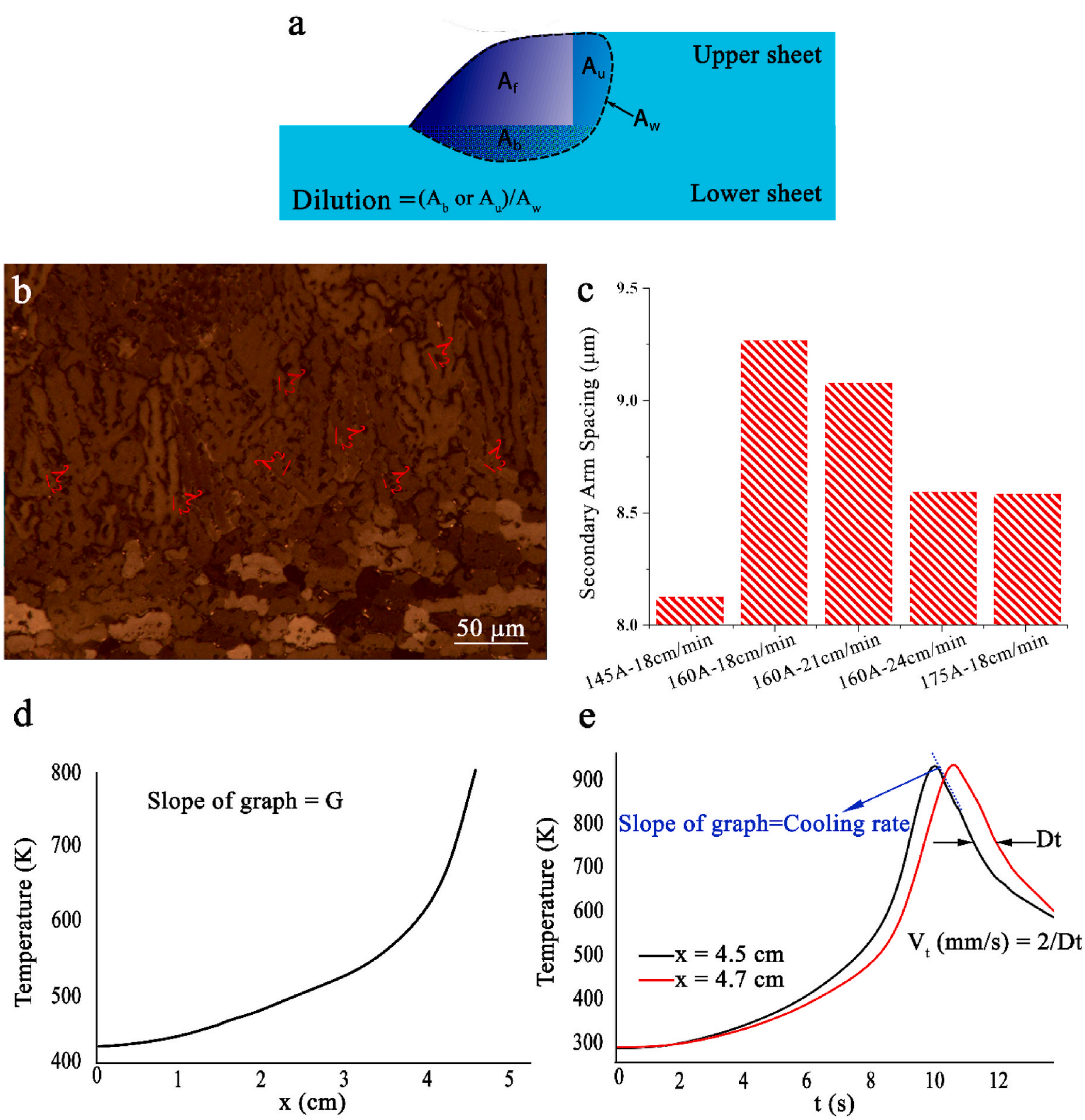


Fig. 5. a) Schematic of dilution calculation method in welds, b) Typical microstructure of the weld and measurement method of secondary arm spacing, c) Measured values for the secondary arm spacing of dendrites d) Temperature-x graph of sample welded with 145 A −18 cm/min, showing weld temperature gradient and e) Temperature-time graph of sample welded with 145 A −18 cm/min, showing cooling rate and isotherm velocity.

Table 3
Chemical composition of the welds (in wt.%) given the weld dilution and chemical compositions of the base and filler alloys presented in Table 1.

Welding parameters	Si	Cu	Mg	Mn	Fe	Cr	Zn	Ti	Ni
145A-18 cm/min	0.39	1.29	2.82	0.5	0.29	0.15	0.15	0.25	0.13
160A-18 cm/min	0.42	1.85	2.16	0.44	0.3	0.13	0.14	0.3	0.23
160A-21 cm/min	0.43	1.57	2.29	0.43	0.29	0.14	0.14	0.29	0.18
160A-24 cm/min	0.39	1.81	2.52	0.51	0.29	0.12	0.14	0.28	0.22
175A-18 cm/min	0.4	2.4	2.02	0.48	0.31	0.1	0.13	0.33	0.32

during the weldability test, the lower base sheet moves at a constant speed. Therefore, the low-temperature base metal repeatedly reaches the weld line, which reduces the melting of the lower base sheet and decreases the contribution of the lower base and the copper (Cu) content in the weld metal. It is of importance that in the welding simulation and SCS model, the movement of the lower base sheet is not considered. Therefore, the calculated copper content in the weld metal is higher compared to the copper content of the weld metal in the weldability test.

5. Conclusion

Dissimilar gas tungsten arc welding of Al–Cu–Mg to Al–Mg–Si aluminum alloys with an Al–Mg filler metal and in a lap joint configuration was simulated using the Flow-3D software. The outcomes of the simulation, conducted at a welding current of 145–175 A and welding speed of 18–24 cm/min, were used as inputs for the solidification cracking model to determine the Solidification Cracking Susceptibility

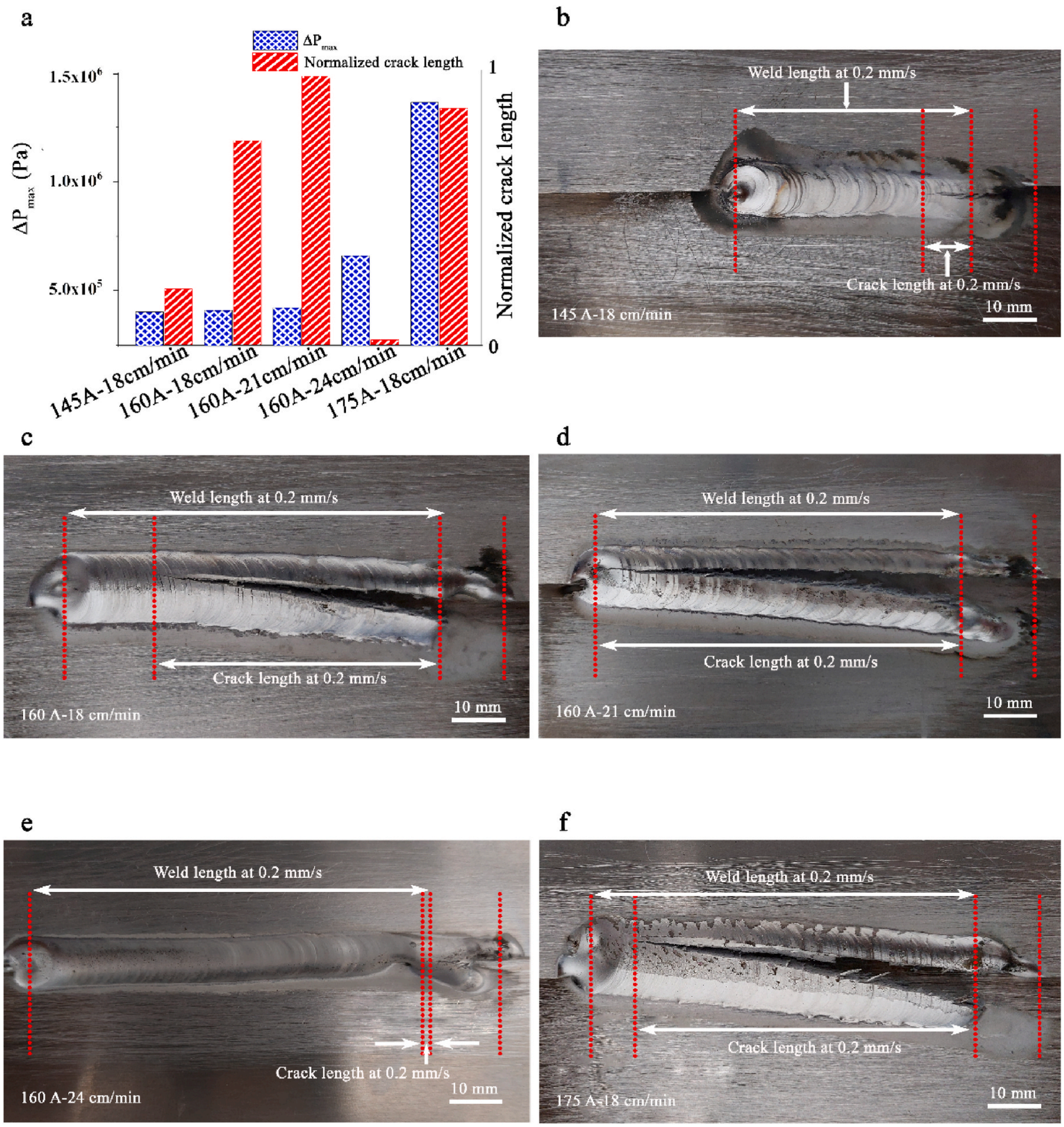


Fig. 6. a) ΔP_{max} (directly related to the solidification cracking susceptibility) for various welds obtained from RDG model, b-f) Weldability test results indicating the weld and crack lengths at various welding parameters.

(SCS) of welds. The following main results were obtained from the work:

- Generally, the weld profiles and maximum temperature at a specified point, obtained from the simulation showed good agreement with the experimental observations. Nonetheless, there was a disparity of 10–23% noted between the experimental and simulated results relating to a specific dimension of the weld profile—the distance between the fillet weld corners. Efforts were made to identify the underlying reasons.
- In the cases studied in this work, the prediction of the weld solidification cracking susceptibility showed that with the increase in welding current from 145 to 175 A at the welding speed of 18 cm/min, the susceptibility to solidification cracking increased by about 2.4 times. This finding was in full agreement with the results of the weldability test, i.e. normalized crack length. The higher susceptibility to solidification cracking was due to the higher melt pool viscosity caused by increased dilution of the Al–Cu base alloy and the related higher copper content in the weld metal as well as longer liquid channel between the adjacent dendrites.
- The length of the liquid channel between two adjacent dendrites, a primary factor affecting susceptibility to solidification cracking, increases with the higher welding current, thereby making liquid backfilling more difficult. Additionally, a higher content of heavy elements such as Cu in the weld pool also increases the difficulty of backfilling. Both of these factors contribute to an increase in the solidification cracking susceptibility.
- The model predicted higher solidification cracking susceptibility with enhancement of the welding speed. This was associated with the lower secondary arm spacing, higher velocity of isotherms and reduced temperature gradient in the weld pool. While the results of the model and weldability test were in good agreement for low and medium welding speeds, a discrepancy was observed at high welding speeds. The reasons were discussed at the light of mechanical stresses

imposed on the weld metal and the horizontal movement of the lower base sheet during the weldability test.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported by the Iran National Science Foundation (Grant No.97010794).

References

[1] Arul S, Sellamuthu R. Application of a simplified simulation method to the determination of arc efficiency of gas tungsten arc welding (GTAW) and experimental validation. Int J Comput Mater Sci Surf Eng 2011;4:265–80.
[2] Vivek Pamarthi V, Sun T, Das A, Franciosa P. Tailoring the weld microstructure to prevent solidification cracking in remote laser welding of AA6005 aluminium alloys using adjustable ringmode beam. J Mater Res Technol 2023;25:7154–68.
[3] Tian Z, Chen S, Wang Y, Tao W, Ye X, Ren W, Yang S. Dynamic laser welding hot cracking behavior and mechanism of new structural material Ni e 28W e 6Cr alloy for molten salt reactor. J Mater Res Technol 2023;25:3022–36.
[4] Liu J, Zeng P, Wu Y, Kou S. Determination of tensile strain causing solidification cracking in welding. Sci Technol Weld Join 2020;25:431–7.
[5] Wu L, Cheon J, Kiran DV, Na SJ. CFD simulations of GMA welding of horizontal fillet joints based on coordinate rotation of arc models. J Mater Process Technol 2015;231:221–38.
[6] Lai H, Hsieh H, Kuo C, Wu W. Solidification cracking nature and sequence of different stainless steels. J Mater Res Technol 2023;25:1030–40.
[7] Han L, Santillana B, Zhuang L. Analytical microsegregation model to study the effect of liquidus correlation on hot cracking susceptibility. J Mater Res Technol 2022;16:1588–96.
[8] Rappaz M, Drezet J-M, Gremaud M. A new hot-tearing criterion. Metall Mater Trans 1999;30:449–55.
[9] Kou S. A criterion for cracking during solidification. Acta Mater 2015;88:366–74.
[10] Zhang Z, Wu C. Effect of fluid flow in the weld pool on the numerical simulation accuracy of the thermal field in hybrid welding. J Manuf Process 2015;20:215–23.
[11] Dong B, Cai X, Ni Z, Lin S, Fan C, Yang C. Numerical simulation of arc characteristics in narrow gap TIG welding. Int J Mech Sci 2019;161–162:105031.
[12] Hao H, Gao J, Huang H. Numerical simulation for dynamic behavior of molten pool in tungsten inert gas welding with reserved gap. J Manuf Process 2020;58:11–8.
[13] Fan X, Qin G, Jiang Z, Wang H. Comparative analysis between the laser beam welding and low current pulsed GMA assisted high-power laser welding by numerical simulation. J Mater Res Technol 2022;22:2549–65.
[14] Ke W, Zeng Z, Oliveira JP, Peng B, Shen J, Tan C, Song X, Yan W. Heat transfer and melt flow of keyhole, transition and conduction modes in laser beam oscillating welding. Int J Heat Mass Tran 2023;203:123821.
[15] Li BL, Zhang J, Ye F, Tang R, Dong Q, Chen J. An approach to studying the hot tearing mechanism of alloying elements in ternary Mg-Zn-Al alloys. J Mater Process Technol 2023;317:117980.
[16] Bodaghi F, Movahedi M, Kokabi AH, Tavakoli R. Effect of solid fraction , grain misorientation and grain boundary energy on solidification cracking in weld of Al – Cu aluminum alloys. Mater Res Express 2019;6:086598.
[17] Soysal T, Kou S. A simple test for solidification cracking susceptibility and filler metal effect. Weld J 2017:389–401.
[18] Cho DW, Park JH, Moon HS. A study on molten pool behavior in the one pulse one drop GMAW process using computational fluid dynamics. Int J Heat Mass Tran 2019;139:848–59.
[19] Park K, Jeong H, Baek S, Kim D, Kang M, Cho J. Turbulent molten pool analysis of tandem GMA automotive steel sheet welding. Int J Heat Mass Tran 2019;129:1–6.
[20] Stenbacka N. On arc efficiency in gas tungsten arc welding. Soldagem & Inspeção 2013;18:380–90.
[21] Cheon J, Kiran DV, Na SJ. CFD based visualization of the finger shaped evolution in the gas metal arc welding process. Int J Heat Mass Tran 2016;97:1–14.
[22] Wang Y, Lü X, Jing H. Dynamic simulation of short-circuiting transfer in GMAW based on the "mass-spring" model. Int J Adv Manuf Technol 2016;87:897–907.
[23] Doodman Tipi AR, Hosseini Sani SK, Pariz N. Improving the dynamic metal transfer model of gas metal arc welding (GMAW) process. Int J Adv Manuf Technol 2015; 76:657–68.
[24] Goldak J, Bibby M, Moore J, House R, Patel B. Computer modeling of heat flow in welds. Metall Trans A B 1986;17:587–600.
[25] Coniglio N, Cross CE. Effect of weld travel speed on solidification cracking behavior. Part 1: weld metal characteristics. Int J Adv Manuf Technol 2020;107: 5011–23.